# Sequences, Series, and the Binomial Theorem
## Answer Key
Algebra 2

### Quick Answers

1. 35
2. 36
3. $x^3 + 9x^2 + 27x + 27$
4. 640
5. 64
6. $16x^4 - 32x^3 + 24x^2 - 8x + 1$
7. 135
8. -9, 10
9. 243
10. $a^3 + 6a^2b + 12ab^2 + 8b^3$
11. 80
12. 1

### Common wrong answers

2. *If they got 72: counted ordered pairs, computing the permutation instead of the combination*
4. *If they got 10: reported the binomial coefficient alone and never raised the constant to its power*
5. *If they got 12: doubled the row number instead of raising two to it*
7. *If they got 45: left the constant unraised, multiplying the coefficient by three instead of by three squared*
9. *If they got 32: used the plain row sum for a binomial whose second term is not one*
11. *If they got 20: kept the two inside the first term unraised, so only the coefficient was used*
12. *If they got 625: ignored the minus sign and added the two constants before raising*

## 1. Evaluate $\binom{7}{3}$.

**Step 1 (choose the tool).** A binomial coefficient counts unordered selections, so use $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ with $n = 7$, $r = 3$. **Step 2 (cancel first).** $\dfrac{7!}{3!\,4!}$: cancel the 4! against the tail of 7!, leaving $\dfrac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1}$. **Step 3 (multiply).** $\dfrac{210}{6} = \boxed{35}$. Cancelling before multiplying keeps the arithmetic small enough to do without a calculator.

## 2. Evaluate $\binom{9}{2}$; a classmate answers 72.

**Step 1 (name the classmate's count).** 72 is the number of *ordered* pairs that can be drawn from 9 objects — the permutation count, in which choosing A then B is different from choosing B then A. **Step 2 (correct the count).** A binomial coefficient counts unordered selections, so every pair has been counted twice; divide by $2! = 2$. **Step 3 (compute).** $\dbinom{9}{2} = \dfrac{9!}{2!\,7!} = \dfrac{9 \cdot 8}{2} = \boxed{36}$.

**3. Expand $(x+3)^3$.**
**Step 1 (set up the row).** With $n=3$ the coefficients are $1,3,3,1$, and the terms run $x^3$, $x^2$, $x^1$, $x^0$ with the matching powers of 3. **Step 2 (build each term).** $1\cdot x^3$, $3\cdot x^2\cdot 3$, $3\cdot x\cdot 3^2$, $1\cdot 3^3$. **Step 3 (collect).**

$$\boxed{x^3+9x^2+27x+27}$$

**4. Coefficient of $x^2$ in $(x+4)^5$.**
**Step 1 (find $r$).** The general term is $\binom{5}{r}x^{5-r}4^r$; the power of $x$ is $5-r$, so $x^2$ needs $r=3$. **Step 2 (evaluate both pieces).** $\dbinom{5}{3}=10$ and $4^3=64$ — the constant's power is what most answers leave out. **Step 3 (multiply).** $10\cdot 64=\boxed{640}$. Reporting 10 alone would give the count of terms of that shape, not the coefficient.

**5. Sum of row 6 of Pascal's triangle.**
**Step 1 (write the row as a sum).** The row is $\sum_{r=0}^{6}\binom{6}{r}$. **Step 2 (use the theorem).** Substituting $p=1$ and $q=1$ into $(p+q)^6=\sum\binom{6}{r}p^{6-r}q^r$ turns every power into 1, so the expansion collapses to exactly that sum of coefficients, and the left side becomes $2^6$. **Step 3 (compute).** $2^6=\boxed{64}$; adding the entries $1+6+15+20+15+6+1$ gives 64 as a check.

**6. Expand $(2x-1)^4$.**
**Step 1 (identify $p$ and $q$).** Here $p=2x$ and $q=-1$, so both the 2 and the minus sign travel with their factor through every term. **Step 2 (build the terms).** Coefficients $1,4,6,4,1$ multiply $(2x)^4$, $(2x)^3(-1)$, $(2x)^2(-1)^2$, $(2x)(-1)^3$, $(-1)^4$, giving $16x^4$, $-32x^3$, $24x^2$, $-8x$, 1. **Step 3 (collect).**

$$\boxed{16x^4-32x^3+24x^2-8x+1}$$

The signs alternate because $q$ is negative and its power climbs by one each term.

**7. Coefficient of $y^4$ in $(3+y)^6$.**
**Step 1 (find $r$).** With $p=3$ and $q=y$, the general term is $\binom{6}{r}3^{6-r}y^r$, so $y^4$ needs $r=4$. **Step 2 (evaluate both pieces).** $\dbinom{6}{4}=15$, and the exponent on 3 is $6-4=2$, so the constant factor is $3^2=9$. **Step 3 (multiply).** $15\cdot 9=\boxed{135}$. The exponent on 3 comes from the *other* factor of the term: the two exponents must always add to 6.

**8. Solve $\dfrac{n(n-1)}{2}=45$.**
**Step 1 (clear the fraction).** $n(n-1)=90$, so $n^2-n-90=0$. **Step 2 (factor).** $(n-10)(n+9)=0$, giving the two solutions $\boxed{n=10}$ and $\boxed{n=-9}$. **Step 3 (interpret).** A count of teams must be a positive integer, so $n=10$ is the answer to the tournament question; $-9$ solves the equation but not the situation. Check: $\dbinom{10}{2}=45$ ✓.

**9. Sum of the coefficients of $(1+2x)^5$.**
**Step 1 (name the substitution).** The sum of the coefficients of any polynomial in $x$ is its value at $x=1$, because every power of 1 is 1. **Step 2 (substitute).** $(1+2\cdot 1)^5 = 3^5$. **Step 3 (compute).** $3^5 = \boxed{243}$. The familiar $2^n$ row sum is only the special case where the second term is 1; here it is 2, so the base is 3.

**10. Expand $(a+2b)^3$.**
**Step 1 (identify $p$ and $q$).** $p=a$ and $q=2b$, so each term carries a power of 2 as well as a power of $b$. **Step 2 (build the terms).** Coefficients $1,3,3,1$ multiply $a^3$, $a^2(2b)$, $a(2b)^2$, $(2b)^3$, giving $a^3$, $6a^2b$, $12ab^2$, $8b^3$. **Step 3 (collect).**

$$\boxed{a^3 + 6a^2b + 12ab^2 + 8b^3}$$

**11. Coefficient of $x^3y^2$ in $(2x+y)^5$.**
**Step 1 (find $r$).** The general term is $\binom{5}{r}(2x)^{5-r}y^r$, and $y^2$ needs $r=2$, which leaves $(2x)^3$. **Step 2 (evaluate both pieces).** $\displaystyle\binom{5}{2} = 10$ and $(2x)^3 = 8x^3$, so the numeric factor is $10\cdot 8$. **Step 3 (multiply).** The coefficient is $\boxed{80}$. It is not the Pascal's-triangle entry 10, because the first term of the binomial is $2x$, not $x$, and that 2 is raised to the third power along with $x$.

**12. Synthesis: sum of the coefficients of $(3x-2)^4$.**
**Step 1 (substitute).** Evaluate the polynomial at $x=1$: $(3\cdot 1-2)^4 = (3-2)^4$. **Step 2 (compute).** $(3-2)^4 = 1^4 = \boxed{1}$. **Step 3 (interpret).** The sum of the coefficients *is* the value of the polynomial at $x=1$, so this expansion is worth 1 there. The individual coefficients are large — $81$, $-216$, $216$, $-96$, $16$ — but they alternate in sign, so they very nearly cancel. Ignoring the minus sign and computing $(3+2)^4 = 625$ answers a different question entirely: it is the sum of the absolute values of the coefficients.